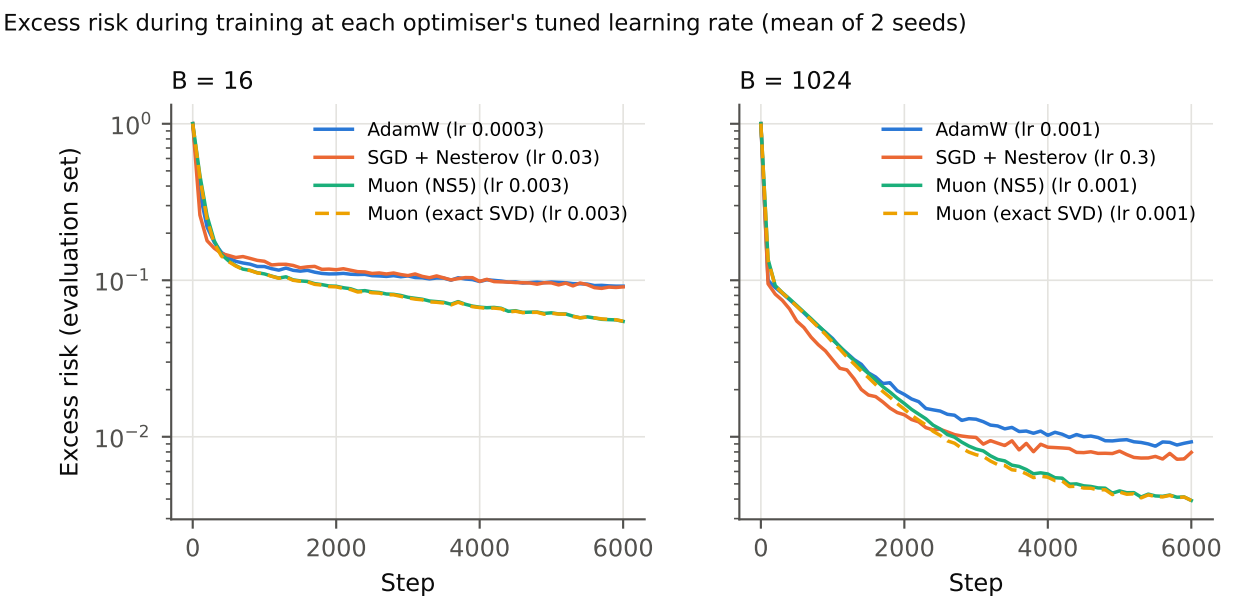
Excess risk during training at each optimiser's tuned learning rate (mean of 2 seeds)
B = 16
B = 1024
Excess risk (evaluation set)
10^0
10^{-1}
10^{-2}
AdamW (lr 0.0003)
SGD + Nesterov (lr 0.03)
Muon (NS5) (lr 0.003)
Muon (exact SVD) (lr 0.003)
AdamW (lr 0.001)
SGD + Nesterov (lr 0.3)
Muon (NS5) (lr 0.001)
Muon (exact SVD) (lr 0.001)
0
2000
4000
6000
Step